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ENTERPRISE SYSTEMS DESIGN AND MODELING

TECHNICAL REPORT TOPIC:

**RISKS FROM AI, IOT INTEGRATION, AND DATA GOVERNANCE IN
GLOBAL ENTERPRISES**

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RISKS FROM AI, IOT INTEGRATION, AND DATA GOVERNANCE IN GLOBAL ENTERPRISES

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EXECUTIVE SUMMARY

This paper focuses on the risks and challenges that are related with AI and IoT integration, while highlighting the importance of data governance in global enterprises. The paper will begin by explaining how these technologies function in global enterprises, with suitable examples from sectors such as manufacturing and logistics. It will then list the advantages and disadvantages of these technologies when applied in terms of cost, energy consumption, human workforce, and more. The paper will also describe the challenges faced by global enterprises, along with future trends and suggest some recommendations for these businesses to improve more on their AI-IoT integration. Lastly, this paper will conclude with how AI-IoT integration represents global enterprise while still highlighting the importance of data governance.

1.0 INTRODUCTION

Technologies like Artificial Intelligence (AI) and the Internet of Things (IoT) are growing as the key drivers of innovation and competitiveness in global enterprises in this digital transformation are. This is because these technologies increase the efficiency, productivity, and decision-making across numbers of business operations. According to Voitek (2025), productivity can reach a minimum of 40% with the use of AI in business processes and 38% improvement in profitability and efficiency. Furthermore, there are 35% of businesses currently employ AI, while 42% are considering adopting these technologies in the future, recording to reports of IBM Global AI Adoption Index

Meanwhile, IoT enables ordinary objects to become “smart” by connecting them to digital networks. IoT can make decisions on their own without the assistance from humans when they are physically connected to one another in certain arrangements. As noted by Kadian (2025), the enterprise IoT market is expected to grow at a 14% compound annual growth rate (CAGR) through 2030. Moreover, IoT-related software-as-a-service (SaaS) and infrastructure-as-a-service (IaaS) spending is projected to rise at a CAGR of over 20% for the remainder of the decade.

However, most global enterprises also should be aware of the risk of integration of AI, IoT. This is because, nowadays there is a huge amount of data being produced by IoT devices, and AI has increased the concerns of data privacy, cybersecurity, and ethical governance. This highlights the importance of

data governance. Data governance consists of rules, policies, and procedures that ensure data accuracy, security, and responsibility. Therefore, this paper aims to analyze the risks related to AI and IoT integration in global enterprises, with an emphasis on how effective data governance can reduce these risks.

2.0 BACKGROUND / LITERATURE REVIEW

AI and IoT are now important parts of digital transformation in many companies. When used together, IoT devices collect real time data, and AI helps to study and make sense of that data to support better decisions. According to Voitekh (2025), this teamwork between AI and IoT helps organizations improve efficiency, use resources wisely, and save costs through automation and predictive maintenance.

However, as more devices become connected, issues like cybersecurity and data management have become serious concerns. Every IoT device can be a possible target to cyberattacks, which is why strong data governance is so important. Stedman (2024) mentioned that having a solid data governance framework helps keep enterprise data reliable, consistent, and secure. Likewise, Kodumuru et al. (2025) stressed the needs for proper data checking methods to avoid bias or errors in AI results.

In conclusion, AI and IoT can bring many benefits to businesses, but weak data governance can lead to problems such as privacy breaches, ethical concerns, and legal issues. For companies that operate in different countries, having consistent and clear data policies is essential to stay accountable and protect information across different regulations.

3.0 ANALYSIS AND DISCUSSION

In this part, it will analyze in detail about how all the technology works with case examples, advantages and disadvantages of these technologies, and provide some recommendations.

3.1 HOW THE TECHNOLOGY WORKS

According to Avila (2025), AI in the enterprise improves the organization in terms of decision making and talent management. With AI, it allows professionals to directly communicate to the data house within them by using simple commands or prompts. In addition, generative AI enhances ERP systems into something beyond an aggregation of data. By transforming the manual into a system, it enables experts to do knowledge based tasks that require greater precision and attention.

Referring to Li et al. (2024), IoT works by connecting the physical things to electronics, software (service), sensor, mobile, intelligent, and network technology. There are few common types of IoT enterprise architectures. Type I IoT enterprise architecture (Figure 1) integrates IoT into basic information systems in which data is gained from things and put it in the EIS databases. The current system that a company has can be called EIS. Next, Type II IoT EA (Figure 2) is a cloud-based mechanism. It uses cloud storage to put it in the data that is gained from physical things, and then performs cloud computing, and provides services to user devices.

Next, based on Holdsworth and Kosinski (2025), data governance function is to ensure data quality, consistency and security across an organization. It requires policies, procedures and standards for

managing the availability, usability, integrity and security of data. Therefore, a data governance program is needed in order to achieve the data governance goals. The data governance program includes various roles with different responsibilities, referring to Figure 3 (Stedman, 2024). They will define a specific goal or set of goals, such as enhancing data quality, supporting compliance or enabling data-driven decision-making. They also select metrics to measure progress toward these goals.

3.2 CASE EXAMPLE OR APPLICATIONS

The use of AI and IoT technologies has changed how industries work, especially in manufacturing and logistics. In manufacturing, many companies use IoT sensors to keep track of their machines and collect data about how they run. Then, AI is applied to study the data and predict if a machine might break down. This allows workers to fix problems earlier and avoid long downtime. For example, a car manufacturing company used an AI and IoT monitoring system and managed to reduce machine breakdowns by 25% (Kodumuru et al., 2025).

In the logistic industry, companies use IoT tracking devices to monitor where their vehicles are and check the condition of goods while being delivered. AI then analyzes the data to plan better delivery routes, improve timing, and make the whole process more efficient. However, since these systems collect a lot of data and may share it between different countries, companies need strong data governance to protect privacy and make sure everything follows the rules (Stedman, 2024). Overall, these examples show that AI and IoT can help businesses work faster and smarter.

3.3 ADVANTAGE AND DISADVANTAGE

Listing the advantages and disadvantages of AI, IoT integration and data governance in global enterprise.

Advantage	Disadvantage
IoT produced a real-time data visibility (Li et al., 2024)	Short number of AI expertise in enterprise (Voitekh, 2025)
AI increases the operational efficiency, automates repetitive jobs, and streamline procedures. (Voitekh, 2025)	The complexity of the management and process representation is increasing as the number of devices connected to the enterprise increases. (Li et al., 2024)
Generative AI able to save time-consuming tasks from employees onto the system. (Avila, 2025)	Limitations of current IT and IoT governance frameworks. (Sedrati et al., 2023)
Data governance ensures data quality, consistency, security across an enterprise, and gives a single source of truth (SSOT) (Holdsworth & Kosinski, 2025)	Without data governance, it restricts efforts to comply with regulations such as General Data Protection Regulation (GDPR) and Privacy Act. (Stedman, 2024)
With data governance, it ensures that data is consistent and trustworthy and doesn't get misused. (Stedman, 2024)	Numerous devices in the IoT required energy. Problems like the battery problem of continuous sensing that use contextual data and the data of the

	confidence levels of the machine learning model (Li et al., 2024)
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3.4 CHALLENGES AND FUTURE TRENDS

Despite its potential, the integration of AI and IoT in enterprises have several challenges. The significant issue is scalability because devices from many suppliers may not interact well, making it difficult to manage and secure thousands of connections. Furthermore, ensuring ethical accountability in AI-driven decisions is important in order to maintain transparency and public trust (Stedman, 2024).

Emerging trends indicate that edge AI will play a greater role in future systems. With use of edge AI, it could minimize latency and provide a strong privacy protection as by processing data closer to the source. Additionally, the advancement of explainable AI will enable enterprises to support automated decisions and make algorithmic processes more transparent. Global policy alignment will also be anticipated, as regulatory bodies move toward harmonized governance standards to support cross-border data management (Kodumuru et al., 2025).

4.0 RECOMMENDATIONS

Based on the analysis, several strategic recommendations are proposed for global enterprises integrating AI and IoT technologies :

- Enhance Data Transparency and Model Explainability. Documentation such as data sources, processing methods, and AI decision making procedures must be organized. This practice influences accountability and supports compliance with international data and AI regulations. (Stedman, 2024)
- Invest in Training and Compliance Monitoring. This is because professional development should be continuous in order to make sure that staff comprehend AI ethics, IoT security, and data governance principles. Automated compliance tools should also be implemented to detect irregularities and maintain adherence to standards across multiple regions. (Voitekh, 2025).
- Propose a new governance framework for the Internet of Things (IoT), called IoT-Gov. It is a framework with three layers which are Strategic Objectives, Design and Modeling, Implementation (Sedrati et al., 2023)

By implementing these recommendations, enterprises can use the full potential of AI-IoT integration while minimizing security and governance risks.

5.0 CONCLUSION

The combination of AI and IoT has changed how global companies operate. These technologies bring many benefits such as automation, higher efficiency, and new ways to innovate. At the same time, they also create serious concerns about privacy, ethics, and legal compliance. Therefore, the effectiveness of AI-IoT systems mostly depends on the presence of a strong and flexible data governance framework.

Enterprises need to focus on being transparent, accountable, and secure while also keeping up with new technologies like edge AI and federated learning. As data laws continue to change around the world, companies should build governance systems that can work across different countries. Lastly, the successful implementation of AI and IoT will not only come from how responsibility data is managed and how well governance rules are followed to build trust and long term sustainability.

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APPENDIX / LOGBOOK

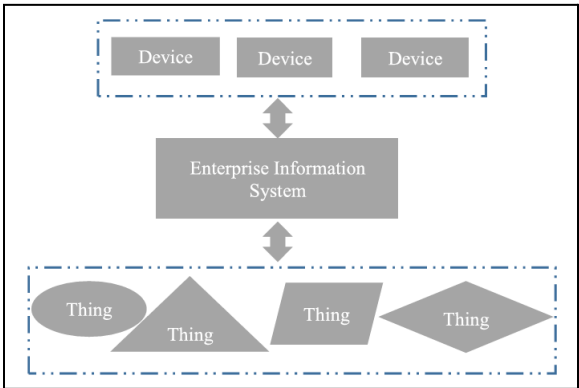


Figure 1 : Type I Enterprise IoT Architecture

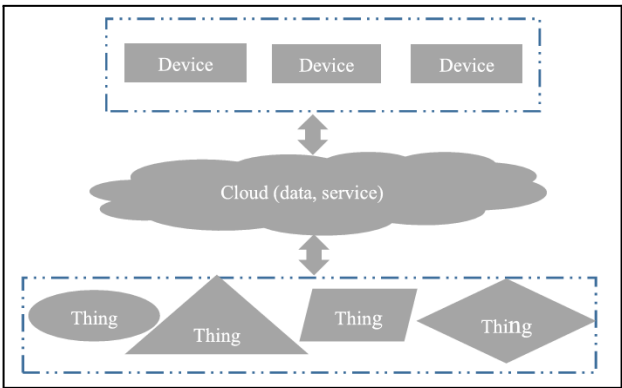


Figure 2 : Type II Enterprise IoT Architecture

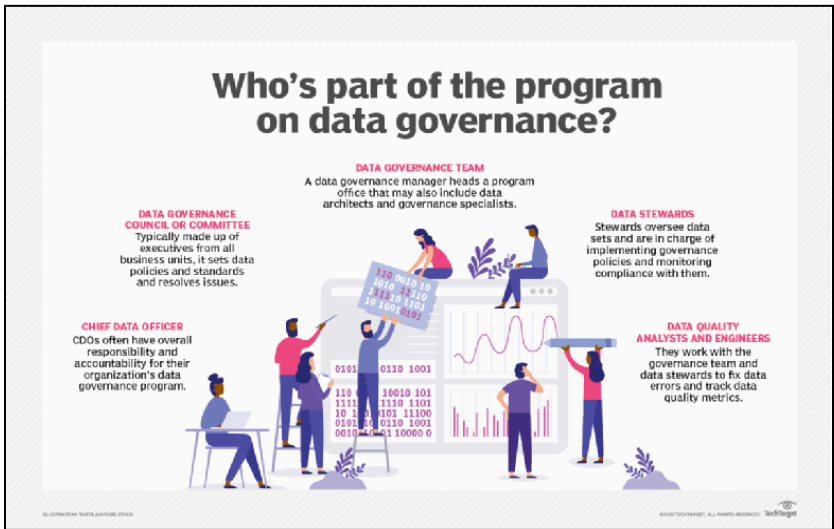


Figure 3 : Data governance process

AI Prompt

Date	Tool/Method	Prompt
20/10/2025	ChatGPT	Explain key risks from IoT integration in enterprises
20/10/2025	ChatGPT	Create outline for technical report on AI, IoT, and data governance